SkillMatch

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Introduction

Project Description

- SkillMatch is a dynamic web application that allows skilled service providers referred to as “workers” to match with potential service seekers referred to as “employers” to connect, meet, and provide/receive a service based on their location.

Design and Implementation

- The Frontend of SkillMatch is developed using **React**, it allows us to create an engaging, responsive and dynamic experience for the users. We utilize Django as it is used as the core backend which provides a framework for development. Then we used...

User Requirements

Requirements Received:

- Both Users (Workers and Employers):
  - Create user profile
  - Connect through a contract
- Worker:
  - View job postings
  - Apply for job
  - Update service status (completed or not)
- Employer:
  - Post Job requests
  - Review/ rate worker's service

Functional Requirements:

- Account creation and user authentication
- Receiving notification when there is a new job request

Non-Functional Requirements:

- Application is easy to navigate for all users
- Application is efficient, load content and respond to user inputs without delay

Requirements Changed: Swapped Employer and Worker requirements

System Requirements

Requirements Received:

- Ability for users to register and create profiles that reflects their role (worker or employer)
- Enable the employer to post a job request.
- System utilizing location data to show the workers who are nearby
- Application is easy to use

Requirements Changed:

- Show the workers based on location for now

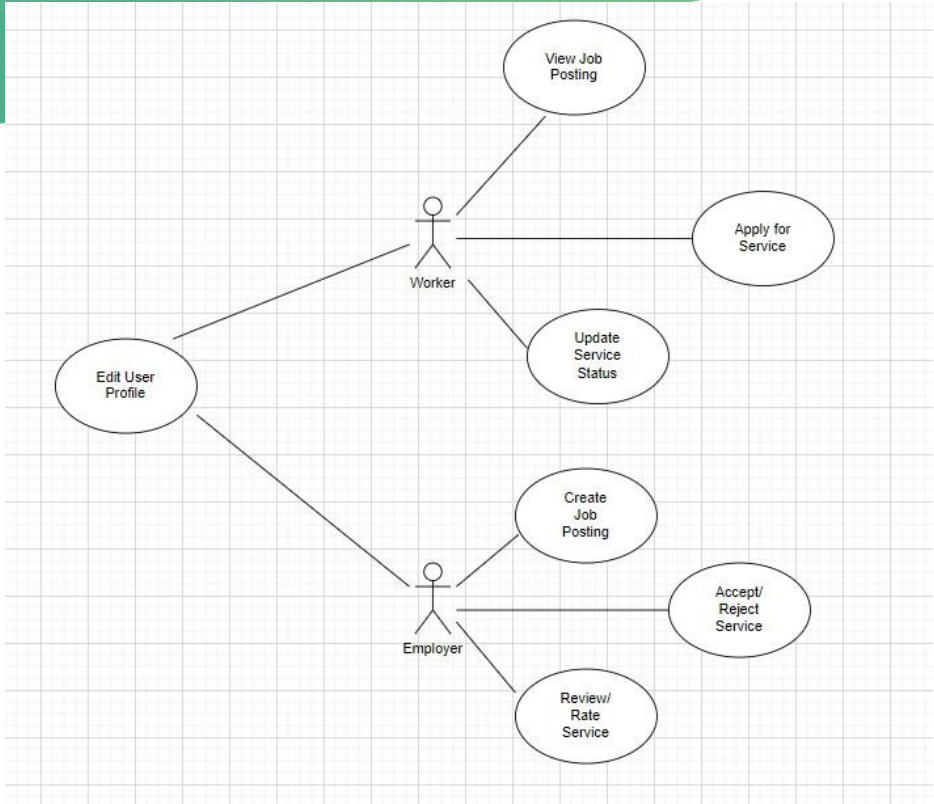
Functional Requirements:

- Enable users to create and manage their profiles
- System shows the workers that are nearby based on location proximity
- Employers can request a job from the workers.
- Review system. Employers are able to leave a rating

Non-Functional Requirements:

- Ensure the application is intuitive and user-friendly
- Achieve the optimal performance under the expected load

Use Case Diagram



Use Case 1: Create Job Posting

Identifier	Use Case #1: Create Job Posting
Purpose	Allow Employers to create job request to find potential workers to complete a service.
Requirements	As a service-seeking user, I want to be able to create job postings to find potential workers to complete a task.
Development Risks	None
Pre-Conditions	Employer is on the job request page.
Post-Conditions	Job posting is visible for the potential workers.

Use Case: Create Job Posting (Typical Course of Action)

Seq#	Actor's Action	System's Response
1	Employer navigates to the job request page	
2	Employer fills out all required input fields	
3	Employer selects "Post Job" button	
4		The system will save job posting information in the database
5		The system displays the job's information on worker's dashboard page

Use Case: Create Job Posting (Alternate Course of Action)

Seq#	Actor's Action	System's Response
1	Employer navigates to the job request page	
2	Employer fills out some of the required input fields	
3	Employer selects "Post Job" button	
4		The system will not save job posting information in the database
5	Employer fills out the rest of the required input fields	
6	Employer selects "Post Job" button	
7		The system will save job posting information in the database and displays it on the worker's dashboard

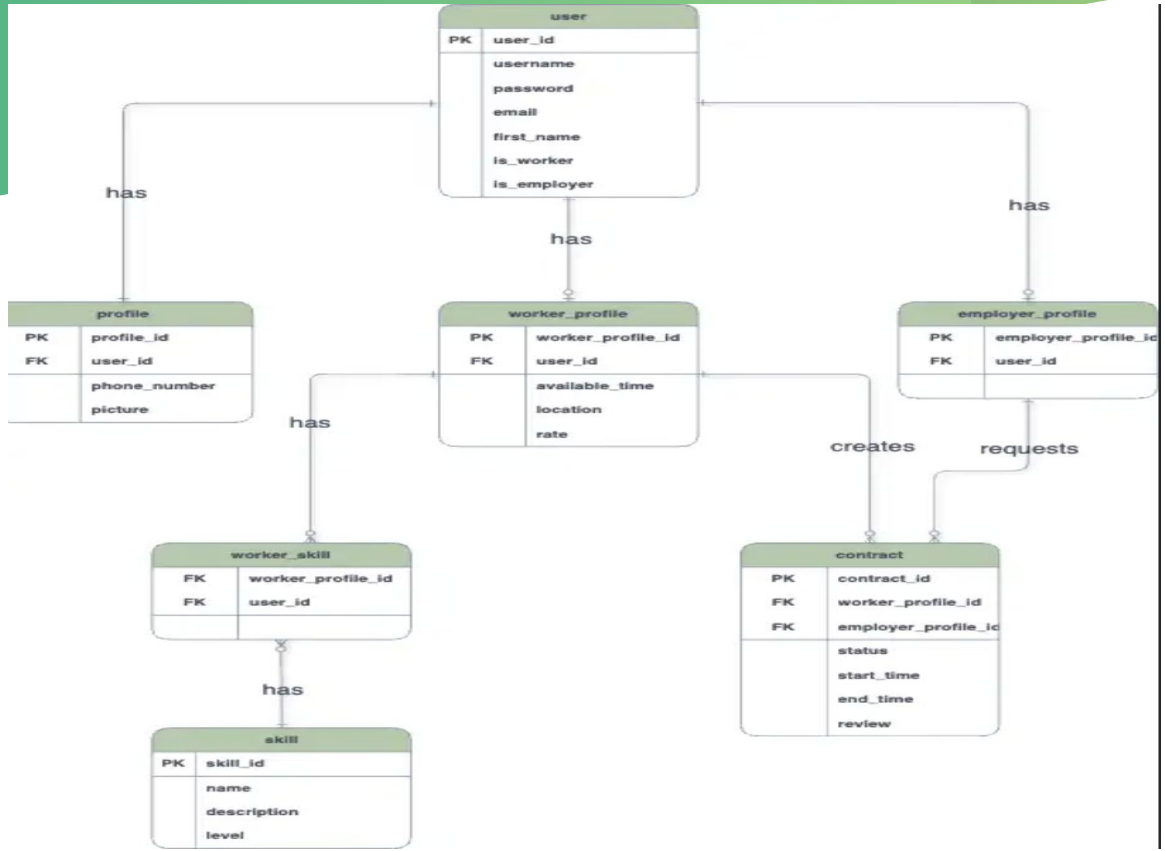
Use Case: Create Job Posting (Exceptional Course of Action)

Seq#	Actor's Action	System's Response
1	Employer navigates to the job request page	
2	Employer doesn't fill out all required input fields	
3	Employer selects "Post Job" button	
4		The system will not save job posting information in the database
5		The system doesn't displays the job's information on worker's dashboard page

Architecture

- Backend Architecture:
  - Django handles logic and data management.
  - REST Framework for building scalable APIs.
- Frontend Architecture:
  - React components for modular and dynamic UI.
  - Bootstrap for responsive styling.
- Database Architecture:
  - PostgreSQL database with tables for users, services,

ER Diagram


user
PK user_id
username
password
email
first_name
is_worker
is_employer
has
has
has
profile
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phone_number
picture
worker_profile
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FK user_id
available_time
location
rate
employer_profile
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creates
requests
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end_time
review
has
skill
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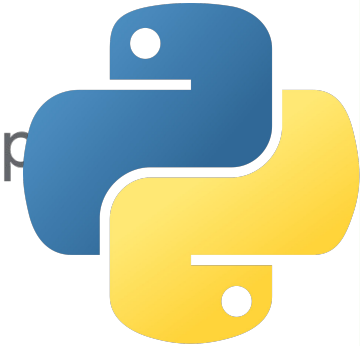
name
description
level

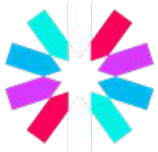
Tech Stack

Map



GitHub

PostgreSQL



JWT

Implementation

Process

- Frontend: HTML, CSS, JSX
- Backend: Python
- IDE: Visual Studio Code

Software

- Tools: Git, Github,
- Framework: React, Django

Deployment

- Locally hosted

Testing Strategy

What we tested

- API Endpoints
- HTTP Request
- Networks

POSTMAN

Demo